

Demand and Supply of Construction Materials in Ramechhap District

In September 2016 the Government of Nepal distributed the first installment of Rs. 50,000 to the households whose houses were damaged by the April/ May 2015 earthquakes. Around 40,000 (95%) households in Ramechhap received the first installment. It was anticipated that this would stimulate construction of houses in the district which would in turn have an impact on the demand of construction materials. It was not clear what would be the impact of this on the demand, quality, price and timely availability of construction materials.

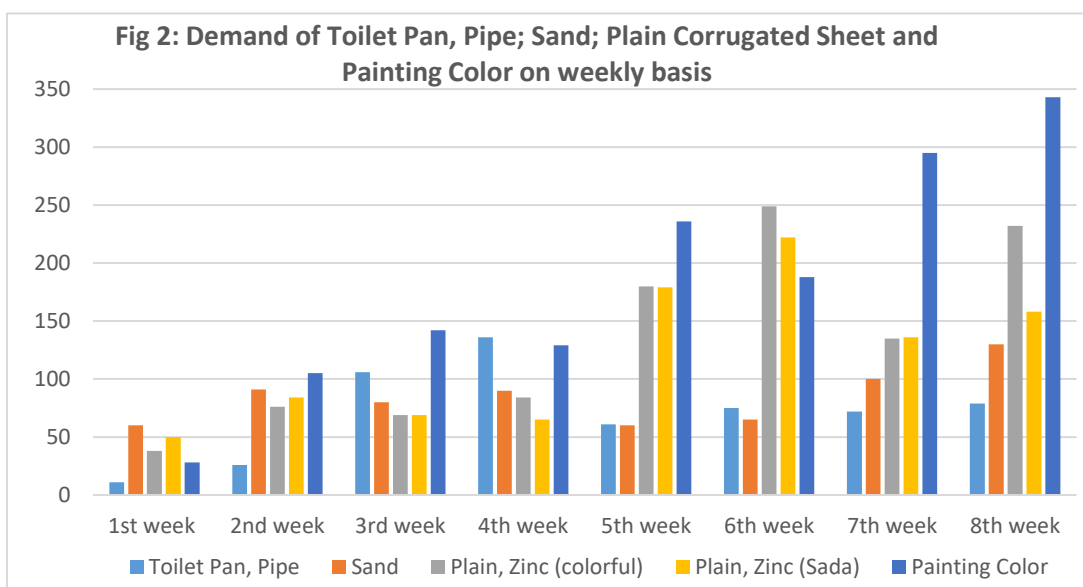
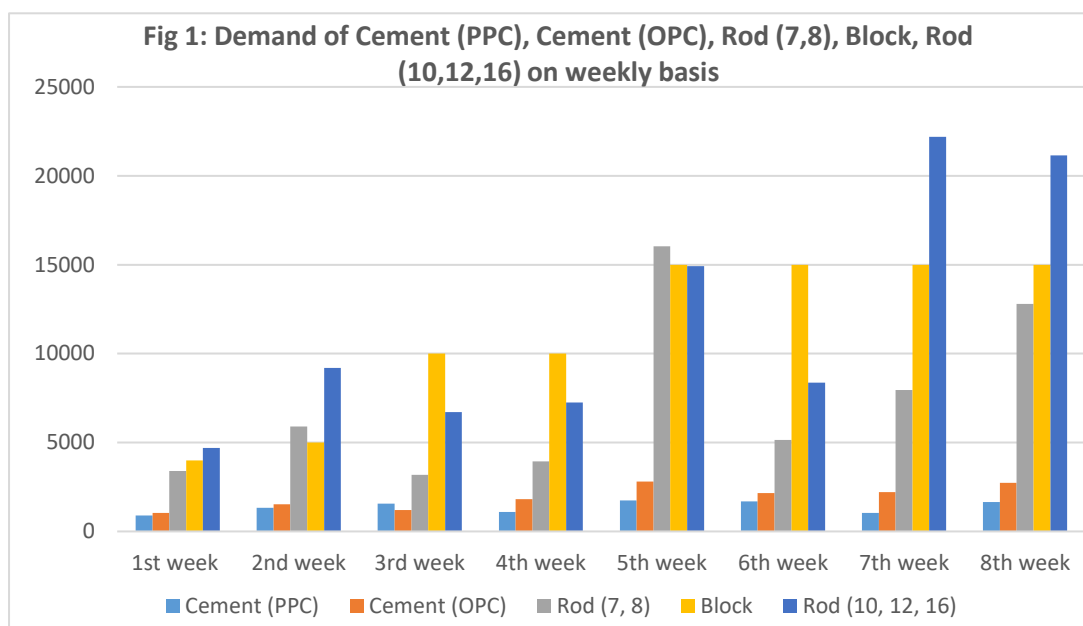
On behalf of the **Swiss Agency for Development and Cooperation (SDC)**, Nepal Agricultural Market Development Programme (**NAMDP**) monitored the price of key construction material such as cement, reinforcement steel bars, sand, aggregates, bricks, concrete blocks, stone, timber and CGI sheets in the district Headquarter Manthali and three other key market centers in the district – Ramechhap, Bhatauli and Khimti. This was done on a weekly basis from the first week of December for eight weeks.

1. Demand and supply of construction materials

There was a steady growth in demand for most of the construction materials monitored in Ramechhap district during the eight weeks of January 2016 and February 2017. Table 1, Figure 1 and Figure 2 show the aggregate demand of construction materials in Ramechhap district over the survey period.

Table 1: Demand of construction material in Ramechhap district on weekly basis

Row Labels	Cement (OPC) (sacks)	Cement (PPC) (sacks)	Rod (7, 8) - Kg	Rod (10, 12, 16) - Kg	Colorful Corrugated sheet (Bundle)	Plain, Corrugated sheet (Bundle)	Toilet Pan, Pipe (set)	Painting Color (liter)	Sand (tractor)	Block (pieces)
1st week	1050	900	3400	4700	38	50	11	28	60	4000
2nd week	1525	1330	5910	9200	76	84	26	105	91	5000
3rd week	1205	1555	3180	6715	69	69	106	142	80	10000
4th week	1820	1095	3938	7247	84	65	136	129	90	10000
5th week	2800	1745	16030	14915	180	179	61	236	60	15000
6th week	2150	1690	5150	8370	249	222	75	188	65	15000
7th week	2220	1040	7960	22200	135	136	72	295	100	15000
8th week	2730	1645	12800	21150	232	158	79	343	130	15000
Grand Total	15500	11000	58368	94497	1063	963	566	1466	676	89000



The market was able to supply the increasing demand of construction materials imported from outside the district (Table 2: Supply of construction material). There was however, a gap in the supply of locally procured materials such as sand and hollow blocks.

Table 2: Supply of construction material on weekly basis

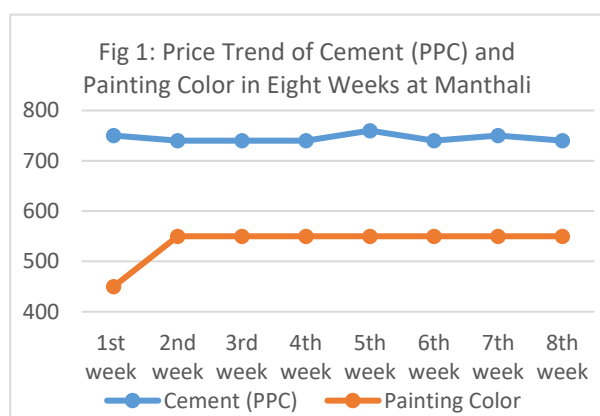
Row Labels	Cement (OPC) - sacks	Cement (PPC) - sacks	Rod (7, 8) Kg	Rod (10, 12, 16) Kg	Colorful Corrugated sheet (Bundle)	Plain corrugated sheet (Bundle)	Toilet Pan, Pipe (set)	Painting Color - Liter	Sand (Tractor)	Block (pieces)
1st week	1050	900	3400	4700	38	50	11	28	60	250
2nd week	1525	1330	5910	9200	76	84	26	105	91	5000
3rd week	1205	1555	3180	6715	69	69	106	142	80	0
4th week	1820	1095	3938	7247	84	97	136	129	90	0
5th week	2800	1745	16030	14915	180	179	61	236	60	0
6th week	2150	1690	5150	8370	249	222	75	188	65	0
7th week	2220	1040	7960	22200	135	136	72	295	50	0
8th week	2730	1645	12800	21150	232	168	79	343	50	0
Grand Total	15500	11000	58368	94497	1063	1005	566	1466	546	5250

2. Price trend of construction materials by market centers

The price of construction materials imported from outside the district remained more or less unchanged in all the market centers of Ramechhap in December 2016 and January 2017. The trend in the price in the main market centers of Ramechhap are given in the sections below.

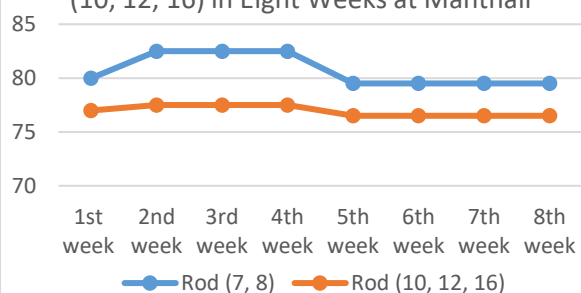
Manthali

The price of OPC Cement and colorful corrugated sheet **remained unchanged** during the eight weeks. The changes in price for the remaining construction materials were also minimal and are shown in the figures below.



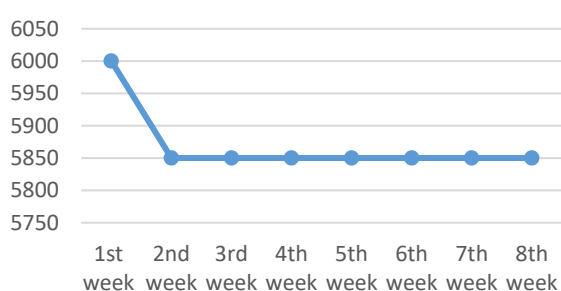
Time Period	Cement (PPC)	Painting Color
1st week	750	450
2nd week	740	550
3rd week	740	550
4th week	740	550
5th week	760	550
6th week	740	550
7th week	750	550
8th week	740	550

Fig 2: Price Trend of Rod (7,8) and Rod (10, 12, 16) in Eight Weeks at Manthali



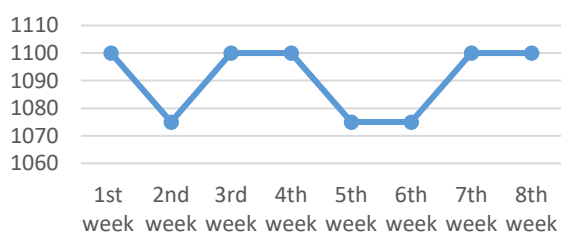
Time Period	Rod (7, 8)	Rod (10, 12, 16)
1st week	80	77
2nd week	83	78
3rd week	83	78
4th week	83	78
5th week	80	77
6th week	80	77
7th week	80	77
8th week	80	77

Fig 3: Price Trend of Plain Corrugated Sheet in Eight Weeks at Manthali



Time Period	Plain Corrugated Sheet
1st week	6000
2nd week	5850
3rd week	5850
4th week	5850
5th week	5850
6th week	5850
7th week	5850
8th week	5850

Fig 4: Price Trend of Toilet Pan, Pipe in Eight Weeks at Manthali

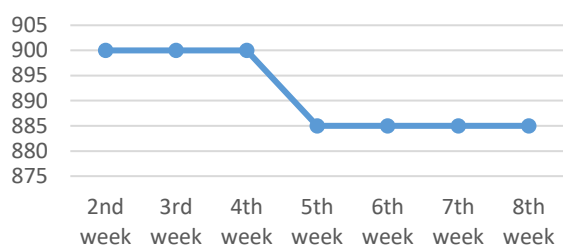


Time Period	Toilet Pan, Pipe (set)
1st week	1100
2nd week	1075
3rd week	1100
4th week	1100
5th week	1075
6th week	1075
7th week	1100
8th week	1100

Ramechhap

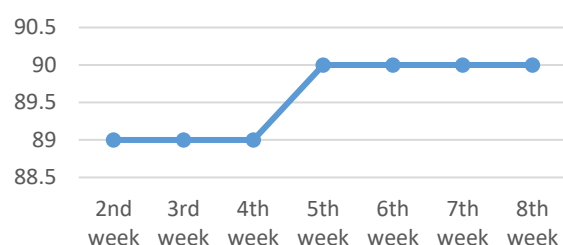
The price of PPC Cement, Rod (10, 12, 16), Toilet Pan set and Painting Color were same in each of the week. The changes in the price for remaining construction materials were *minimum* and are shown in the figures below.

Fig 5: Price Trend of Cement (OPC) in Eight Weeks at Ramechhap



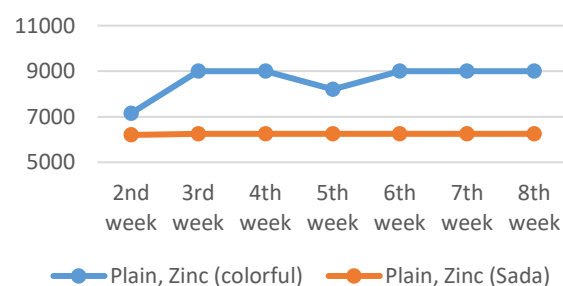
Time Period	Cement (OPC)
2nd week	900
3rd week	900
4th week	900
5th week	885
6th week	885
7th week	885
8th week	885

Fig 6: Price Trend of Rod (7,8) in Eight Weeks at Ramechhap



Time Period	Rod (7, 8)
2nd week	89
3rd week	89
4th week	89
5th week	90
6th week	90
7th week	90
8th week	90

Fig 7: Price Trend of Colorful Corrugated sheet and Plain Corrugated Sheet in Eight Weeks at Ramechhap



Time Period	Colorful Corrugated Sheet	Plain Corrugated Sheet
2nd week	7150	6200
3rd week	9000	6250
4th week	9000	6250
5th week	8200	6250
6th week	9000	6250
7th week	9000	6250
8th week	9000	6250

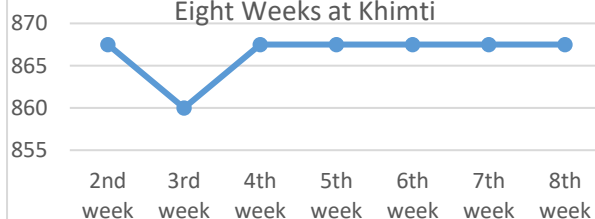
Bhatauli

Prices **remained unchanged** for all the construction materials over the eight-week period.

Khimti

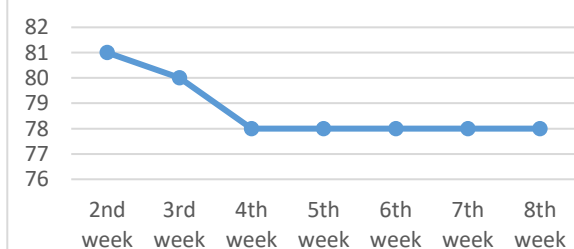
The prices of PPC Cement, Rod (10, 12, 16), Plain Corrugated sheet, Toilet Pan set and Painting Color were same in each of the week. The changes in the price of the remaining construction materials were **minimal** as shown in the figures below.

Fig 8: Price Trend of Cement (OPC) in Eight Weeks at Khimti



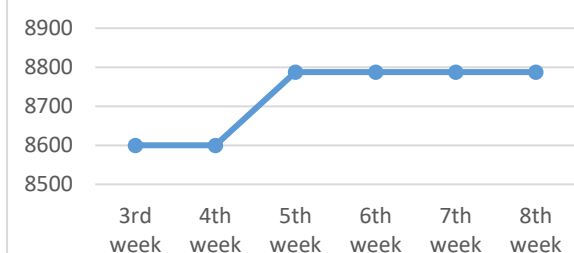
Time period	Cement (OPC)
2nd week	868
3rd week	860
4th week	868
5th week	868
6th week	868
7th week	868
8th week	868

Fig 9: Price Trend of Rod (7,8) in Eight Weeks at Khimti



Time period	Rod (7, 8)
2nd week	81
3rd week	80
4th week	78
5th week	78
6th week	78
7th week	78
8th week	78

Fig 10: Price Trend of Colorful Corrugated Sheet in Eight Weeks at Khimti



Time period	Colorful Corrugated sheet
3rd week	8600
4th week	8600
5th week	8788
6th week	8788
7th week	8788
8th week	8788

Key Observations

1. The demand of all the construction materials in Ramechhap district grew steadily during the monitored period.
2. The increase in demand was matched by increase in supply for all the construction materials imported from outside the district.
3. The supply of locally available construction material such as sand, hollow blocks and wood was lower than the demand.
4. The price of the construction materials imported from outside the district remained stable during the monitored period. Prices of locally available construction material such as sand, hollow blocks and wood increased during the monitored period.
5. The first installment of Rs. 50,000 did not stimulate reconstruction of damaged houses very significantly in the district.

Annex 1: Prices of Construction materials on Weekly Basis in the main market centers of Ramechhap

Time Period	Price of Commodity (Rs.)									
	Cement (OPC) per sack	Cement (PPC) – per sack	Rod (7, 8)- per Kg	Rod (10, 12, 16) per Kg	Colorful Corrugated Sheet (per bundle)	Plain Corrugated Sheet (per bundle)	Toilet Pan, Pipe (per set)	Painting Color (per liter)	Sand (per tractor load)	Block (per piece)
Bhatauli										
1st week	850	750	79	76	8500	6500	800	500	300	65
2nd week	850	750	79	76	8500	6500	800	500	300	65
3rd week	850	750	79	76	8500	6500	800	500	300	
4th week	850	750	79	76	8500	6500	800	500	300	
5th week	850	750	79	76	8500	6500	800	500	300	
6th week	850	750	79	76	8500	6500	800	500	300	
7th week	850	750	79	76		6500	800	500	300	
8th week	850	750	79	76	8500	6500	800	500	300	
Chuchure										
5th week		800	80	77	9200	6600				
Galpa										
6th week	1050	950	86	84	10000	7000	1000	550		
7th week	1025	950	86	84	10000	7000	1000	550		
8th week	1025	950	86	84	10000	7000	1000	550		
Khimti										
2nd week	868	758	81	76				410		
3rd week	860	758	80	76	8600	5500		410		
4th week	868	758	78	76	8600	5500				
5th week	868	758	78	76	8788	5500				
6th week	868	758	78	76	8788	5500				
7th week	868	758	78	76	8788	5500	900	410		
8th week	868	758	78	76	8788	5500	900	410		
Manthali										
1st week	850	750	80	77	8500	6000	1100	450		
2nd week	850	740	83	78	8500	5850	1075	550		
3rd week	850	740	83	78	8500	5850	1100	550		
4th week	850	740	83	78	8500	5850	1100	550		
5th week	850	760	80	77	8500	5850	1075	550		
6th week	850	740	80	77	8500	5850	1075	550		
7th week	850	750	80	77	8500	5850	1100	550		
8th week	850	740	80	77	8500	5850	1100	550		
Ramechhap										
2nd week	900	785	89	78	7150	6200				
3rd week	900	785	89	78	9000	6250				
4th week	900	785	89	78	9000	6250				
5th week	885	785	90	78	8200	6250	900			
6th week	885	785	90	78	9000	6250	900			
7th week	885	785	90	78	9000	6250	900			
8th week	885	785	90	78	9000	6250	900			